

Ransomware and Its Variants

Dr. Ashok Kumar Das

IEEE Senior Member

Professor

Center for Security, Theory and Algorithmic Research
International Institute of Information Technology, Hyderabad

E-mail: *ashok.das@iiit.ac.in*

URL: <https://www.iiit.ac.in/faculty/ashok-kumar-das/>
<https://sites.google.com/view/iitkgpakdas/>

- Ransomware is a type of malicious software that encrypts or locks data.
- Attackers demand payment (ransom) to restore access.
- Common targets: individuals, enterprises, healthcare, and government sectors.

How Ransomware Works

- 1 Infection via phishing emails, malicious links, or exploits.
- 2 Execution of malware on the victim system.
- 3 Encryption or locking of files/system.
- 4 Ransom demand displayed to the victim.

Types of Ransomware

- Crypto Ransomware
- Locker Ransomware
- Scareware
- Doxware (Leakware)
- Ransomware-as-a-Service (RaaS)

- Encrypts user files using strong cryptographic algorithms.
- Files become inaccessible without decryption key.

Example: WannaCry, CryptoLocker

- Locks users out of their systems completely.
- Prevents access to desktop or OS functions.

Example: WinLocker

- Displays fake warnings about infections.
- Tricks users into paying for fake solutions.

Example: Fake antivirus software

Doxware (Leakware)

- Threatens to publish sensitive data if ransom is not paid.
- Targets privacy-sensitive information.

Example: Maze ransomware

Ransomware-as-a-Service (RaaS)

- Cybercriminals offer ransomware kits to others.
- Lowers the barrier for launching attacks.

Example: REvil, DarkSide

Notable Ransomware Attacks

- WannaCry (2017): Global attack exploiting SMB vulnerability.
- Petya/NotPetya: Targeted enterprises and caused massive damage.
- Colonial Pipeline Attack (2021): Disrupted fuel supply in the US.

- AIIMS Delhi Cyberattack (2022): Major ransomware attack on hospital servers.
- Disrupted digital services including patient records and billing systems.
- Attackers reportedly demanded ransom for restoring access to critical data.
- Highlighted vulnerabilities in healthcare cybersecurity infrastructure.

Impact of Ransomware

- Financial losses
- Data breaches
- Operational disruption
- Reputational damage

- Regular data backups
- Keep software updated
- Use antivirus and firewalls
- Avoid suspicious emails/links

- Ransomware is a major cybersecurity threat.
- Understanding its variants helps in better defense.
- Prevention and awareness are key.